

Python Learning Roadmap

Beginner Level

- Introduction to Python: Learn what Python is, how to install it, and run your first 'Hello, World!' program.
- Variables and Data Types: Understand how to store data using variables and explore types like strings, numbers, and booleans.
- Operators: Perform calculations and comparisons using Python's arithmetic, logical, and relational operators.
- Conditional Statements: Control the flow of your program using if-else conditions and logical decision-making.
- Loops: Automate repetitive tasks using `for` and `while` loops along with loop control statements.
- Functions: Write reusable blocks of code with functions, and understand parameters, arguments, and return values.
- Data Structures: Organize data efficiently using lists, tuples, sets, and dictionaries with their basic operations.
- String Manipulation: Explore slicing, formatting, and commonly used string methods for processing text.
- Basic Error Handling: Handle errors gracefully using try-except blocks and understand common Python exceptions.

Intermediate Level

- Modules and Packages: Import built-in or third-party libraries, and structure your code with reusable modules.
- File Handling: Learn how to read from and write to files in different formats using Python.
- Comprehensions: Write concise and readable code with list, dictionary, and set comprehensions.
- Object-Oriented Programming (OOP): Understand classes, objects, inheritance, and how to build real-world models in code.
- Lambda, map, filter, reduce: Use anonymous functions and functional programming tools to process data efficiently.
- Working with JSON: Learn to parse and write JSON data - essential for working with APIs and configs.

- Decorators & Generators: Master advanced functions with decorators and generate sequences using generators and ``yield``.

Advanced Level

- Regular Expressions: Use pattern matching with the ``re`` module to search, extract, and validate strings.
- Working with Databases: Connect Python to SQLite/MySQL and perform basic database operations like insert, update, delete.
- Web Scraping: Extract data from websites using libraries like ``requests`` and ``BeautifulSoup``.
- Multithreading & Multiprocessing: Speed up your programs with concurrency using threads and processes.
- APIs and HTTP Requests: Interact with external services by sending and receiving data using Python's ``requests`` module.
- Unit Testing: Write automated tests for your functions and programs using the ``unittest`` framework.

Optional Special Topics

- GUI with Tkinter / PyQt: Create desktop applications with buttons, forms, and event handling.
- Data Analysis with Pandas, NumPy: Work with large datasets, perform calculations, and clean data efficiently.
- Plotting with Matplotlib / Seaborn: Visualize your data through charts, graphs, and plots.
- Intro to Django/Flask: Build your first web applications with Python-based web frameworks.
- Machine Learning with scikit-learn: Apply basic machine learning techniques using one of Python's most powerful libraries.